

RAMCO Institute of Technology, Rajapalayam
Department of Mechanical Engineering
Academic Year 2020-2021 (Odd Semester)

Innovative Practices(s) Description

Name of the Faculty	:	R.Prabhakaran, AP/Mechanical
Subject Code and Title	:	ME8097 – Non Destructive Testing and Evaluation
Year and Branch	:	IV –Mechanical Engineering
Topics Covered	:	Tensile Test on Mild Steel
Name of the Laboratory	:	Strength of Material Laboratory
Website Link	:	http://sm-nitk.vlabs.ac.in/exp13/index.html
No.of Students participated	:	35
Date & Time	:	24.08.2020 & 10.40AM to 11.40AM

Demonstration of Tensile Test on Mild Steel

OBJECTIVE: To study the mechanical properties of Mild Steel specimen under tension load.

In tension test ends of a test piece are fixed into grips connected to a straining device and to a load measuring device. The test involves straining a test piece by tensile force generally to fracture for the purpose of determining one or more of the mechanical properties.

The students can able to evaluate Yield Stress, Tensile Strength, Modulus of Elasticity, Percentage Elongation and Percentage reduction in area after completion of this virtual laboratory.

OUTCOME : The Virtual Laboratory Experiments helps in attaining the following CO-PO mapping:

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO1 - Explain the fundamental concepts of NDT.	3	2	1	1	3	2	2	2	2

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24 Aug, 2020 | 03:25:28 PM

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An Initiative of
Ministry of Human Resource Development
Under the National Mission on Education
through ICT

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OBJECTIVES THE PHILOSOPHY SALENT FEATURES

Objectives

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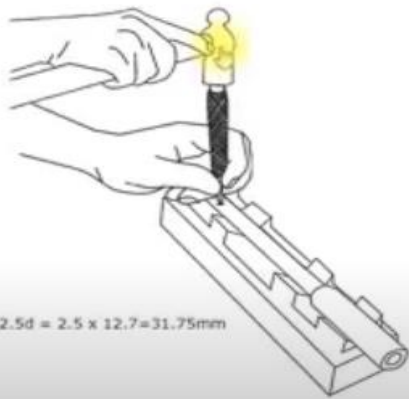
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TENSILE TEST ON MILD STEEL

STEP 3 Punch marks are made at a distance of 2.5d.



Distance $2.5d = 2.5 \times 12.7 = 31.75\text{mm}$

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STEP 11 Observations and Calculations.

Result	Actual Value	Entered Value	Percentage Error
Yield Stress(N/mm^2)	347.52	356	-2.44
Tensile Strength(N/mm^2)	442.3	490	-10.78
Modulus of Elasticity(GPa)	209.65	212	-1.12
Percentage Elongation(%)	29.89	39	-30.48
Percentage reduction in area(%)	67.14	55	18.08

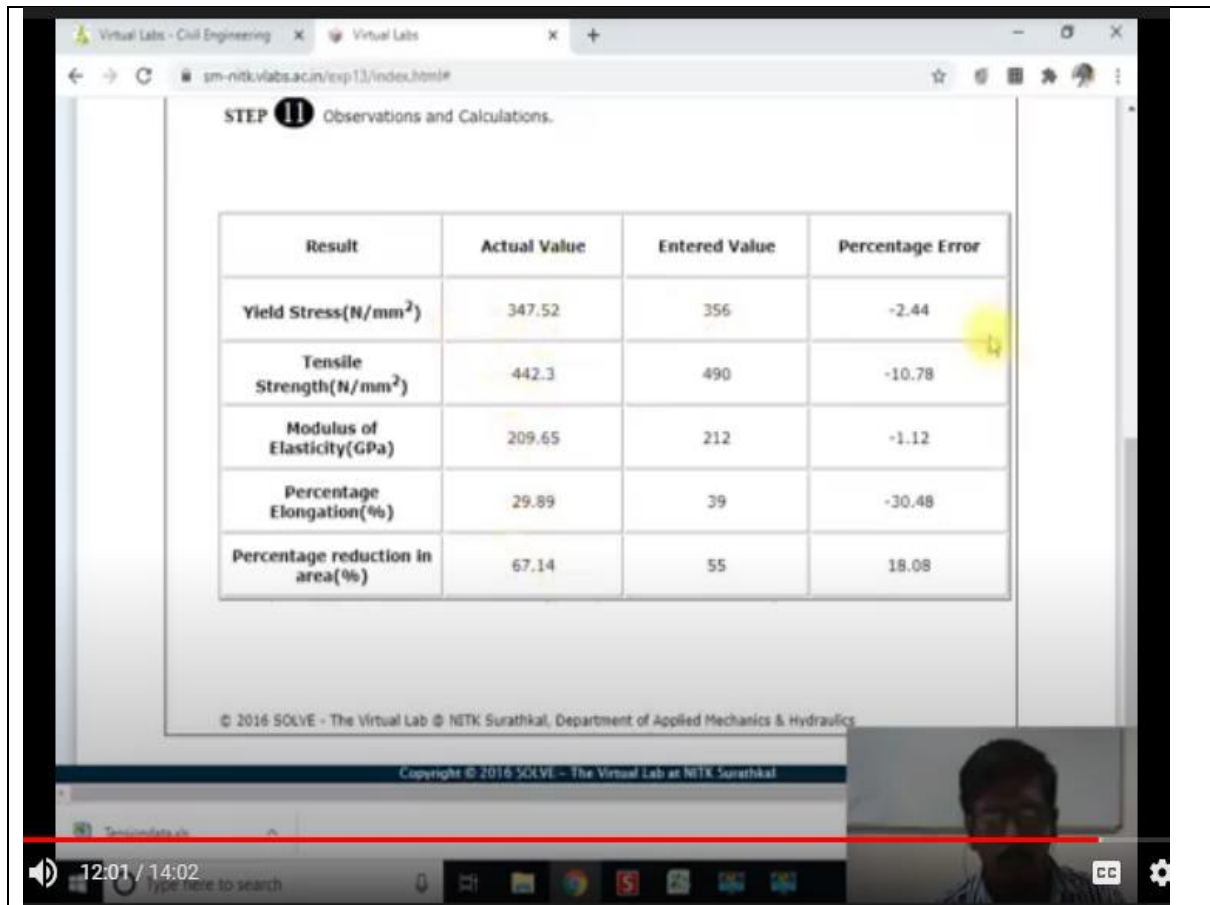
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The image is a screenshot of a web browser displaying a virtual lab interface. The browser's address bar shows 'sm-nitkvlabs.ac.in/exp13/index.html'. The main content area is titled 'STEP 11 Observations and Calculations.' and contains a table with four columns: 'Result', 'Actual Value', 'Entered Value', and 'Percentage Error'. The table lists five mechanical properties: Yield Stress, Tensile Strength, Modulus of Elasticity, Percentage Elongation, and Percentage reduction in area. A yellow highlight is visible on the 'Percentage Error' column for 'Yield Stress'. At the bottom of the page, there is a copyright notice for '© 2016 SOLVE - The Virtual Lab © NITK Surathkal, Department of Applied Mechanics & Hydraulics'. A video player overlay is present at the bottom of the screen, showing a timestamp of '12:01 / 14:02' and a search bar. A small video feed of a person is visible in the bottom right corner of the video player.